Reducing Maintenance with Layouts and Partials

So far in this chapter we've looked at templates as isolated chunks of code and HTML. But one of the driving ideas behind Rails is honoring the DRY principle and eliminating the need for duplication. The average website, though, has lots of duplication:

- Many pages share the same tops, tails, and sidebars.
- Multiple pages may contain the same snippets of rendered HTML (a blog site, for example, may display an article in multiple places).
- The same functionality may appear in multiple places. Many sites have a standard search component or a polling component that appears in most of the sites' sidebars.

Rails provides both layouts and partials that reduce the need for duplication in these three situations.

Layouts

Rails allows you to render pages that are nested inside other rendered pages. Typically this feature is used to put the content from an action within a standard site-wide page frame (title, footer, and sidebar). In fact, if you've been using the `generate` script to create scaffold-based applications, then you've been using these layouts all along.

When Rails honors a request to render a template from within a controller, it actually renders two templates. Obviously, it renders the one you ask for (or the default template named after the action if you don't explicitly render anything). But Rails also tries to find and render a layout template (we'll talk about how it finds the layout in a second). If it finds the layout, it inserts the action-specific output into the HTML produced by the layout.